

Stochastic Differential Equations*

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May 26, 2026

15/05/26

the subject seem to be using a lot of constructs from measure theory and probability theory, the intersection being the probability measure.

16/05/26

i'm reading the papers by song and karras.

17/05/26

20/05/26

i was getting a bit mathsy with rvs and dr. p clearly said to be keep it physicsy and the next week's objective is to understand the fokker planck equation. i have started with the book by prof. v. balakrishnan on elements of non equilibrium statistical mechanics.

26/05/26

i think i've reached a point where i can start writing the exposition.

Langevin equation

The objective of this equation is to model the motion of a particle ("tagged particle" as we call it) in a fluid, which is an elementary problem in the branch of non equilibrium statistical mechanics.

The equation can be systematically obtained from the newton's 2nd law as follows. Consider a tagged particle of mass m in a fluid which is being acted upon by an external force $\mathbf{F}$. Now newton's 2nd law reads,

$$m\dot{\mathbf{v}} = \mathbf{F} \quad (1)$$

Now we split our system into 2 parts, the tagged particle and the rest of the system (i.e., the fluid).

*This is a preliminary draft and currently a work in progress.